

IEEE ASUSB

Computer Arithmetic Workshop

Feb.2026

Task 2

Parallel Prefix Adder

Due date:

Wednesday 11/2/2026

Requirements:

- ✓ **RTL:** the Verilog code implements the assigned question and comment your design well.
- ✓ **Simulation:** create a simple Testbench for the design and run simulation attach screenshots to show the Transcript output and waveform
- ✓ **FPGA:** run synthesis and Implementation using VIVADO Toolchain targeting VCU118 board and report the Utilization and critical path delay calculation, (For Quartus: Use Cyclone V 5CSXFC5D6F31)

[Bonus] Parametrized RTL: write generic code can be used to generate N-bit adder (where N is Parameter/Generic) of type similar to the Type in the Assigned Question

Report: report on the main flaws (if any) in your design and explain the limitations in your generic implementation.

Specs:

You are asked to design a 64-bit parallel prefix adder using Brent-kung Carry network through the following the steps below and attach all the required captures and codes

Fig.3 below show an example how to implement a size-8 prefix sum network using brent-kung Architecture each layer is encoded with different color to help you extract the pattern

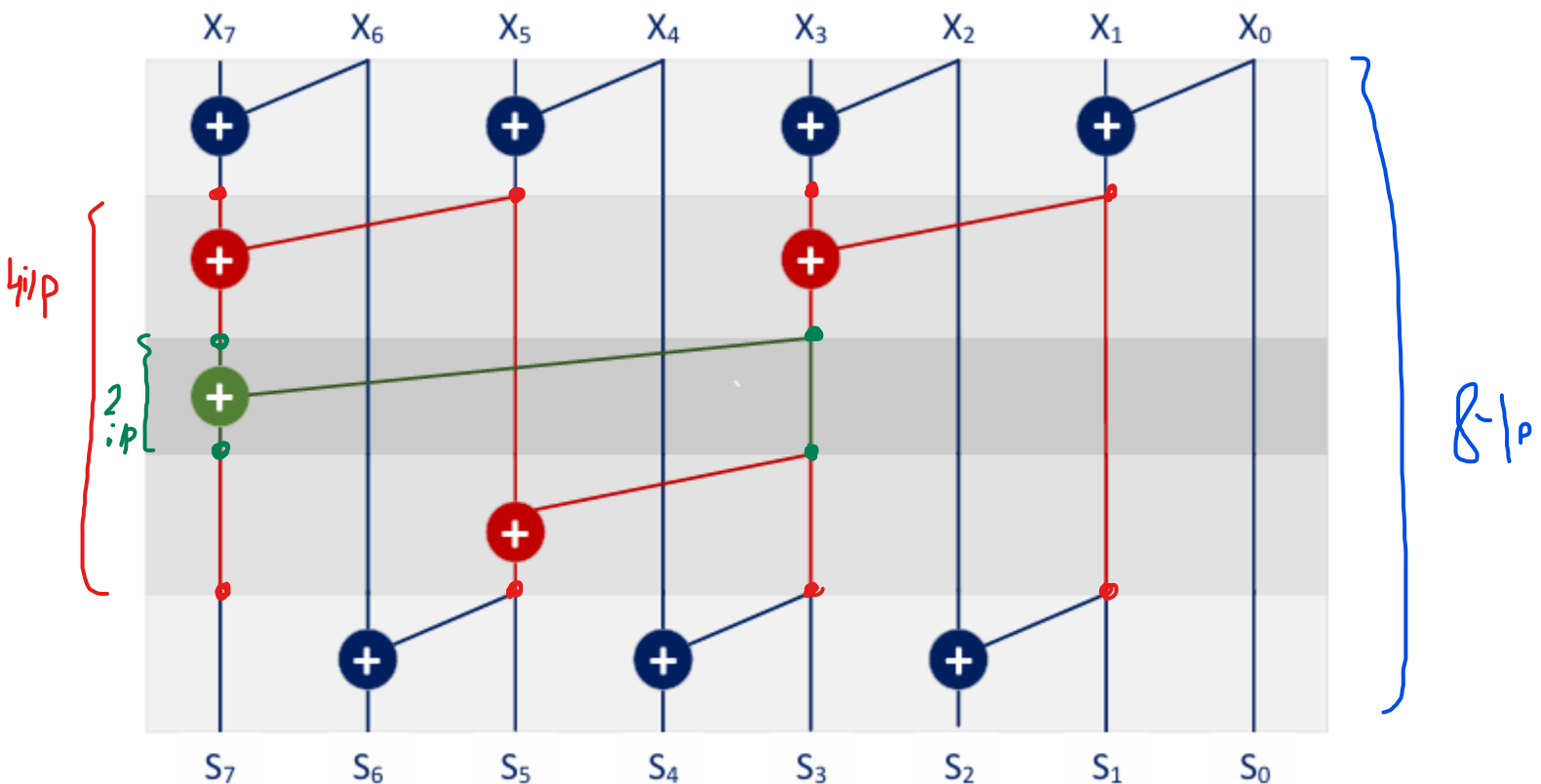


Figure 3

Fig.4 show the steps how the example in fig.3 is build step by step first layer one which contains (8-1 adders) and change the problem from 8-input problem to a smaller one with only 4-inputs then we apply the same idea by using (4-1 adders) and split the problem to half please spend some time to convince yourself and make sure you understand the pattern very well then start to map this into a carry network of 64-bit adder by removing each + sign with a carry operator you will find the implementation of carry operator below in fig.4

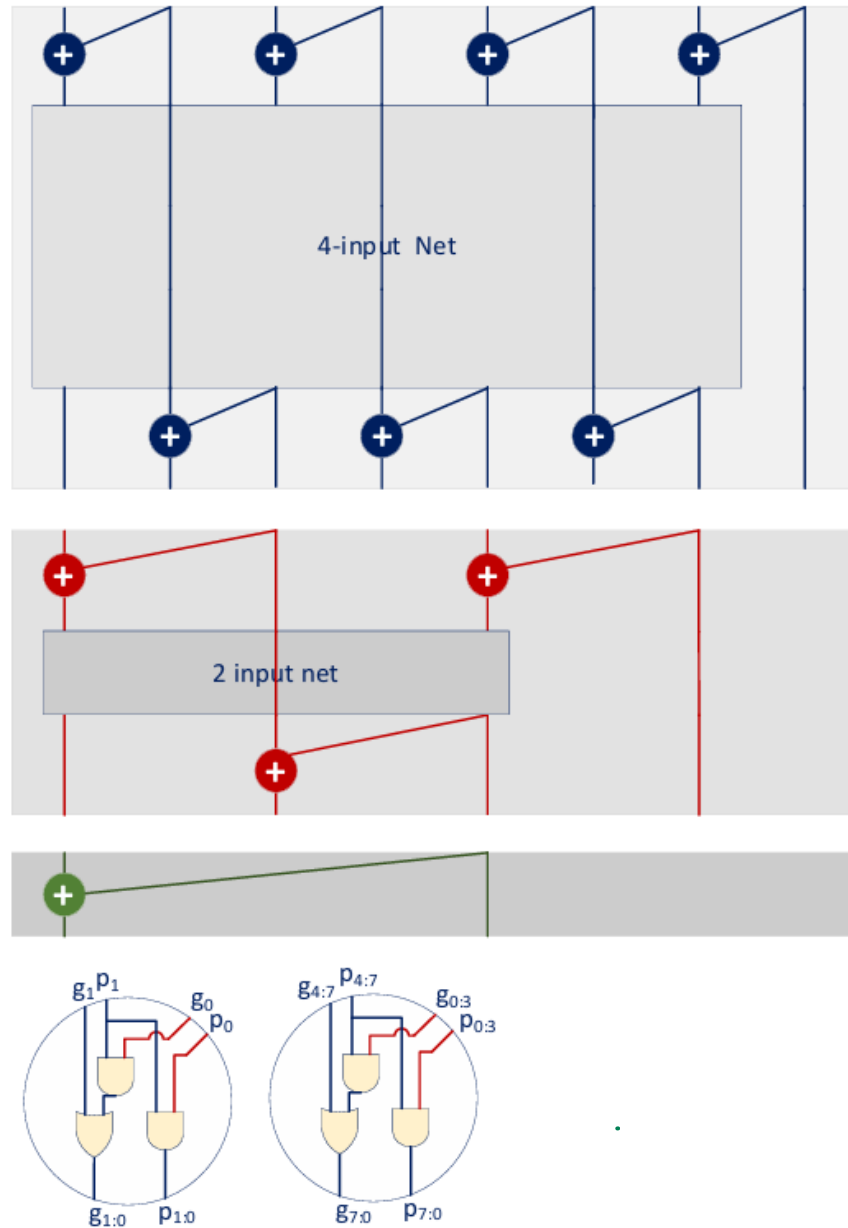


Figure 4

- **Inputs** {A, B, Cin} (A,B are 64-bits inputs and Cin is 1-bit)

- **Outputs** {Sum, Cout} (sum is 64-bits output and Cout is 1-bit)